

7. Create a Jenkins file defining pipeline stages

Create a Jenkinsfile defining pipeline stages (build, test, deploy) and demonstrate a declarative or scripted pipeline execution.

```
ayushi2323@LAPTOP-6LVIDNHT:~$ cd ~/docker-project
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ ^C
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ ls
Dockerfile      README.md      main.py
'LAB2 TASK2.pdf'  commands.txt  observations.txt
'Lab2 (1).docx'   lab2.docx     requirements.txt
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ ls
Dockerfile      README.md      main.py
'LAB2 TASK2.pdf'  commands.txt  observations.txt
'Lab2 (1).docx'   lab2.docx     requirements.txt
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ nano Jenkinsfile

ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ cat Jenkinsfile
pipeline {
    agent any

    stages {
        stage('Build') {
            steps {
                echo 'Building the application...'
                sh 'docker build -t my-docker-app .'
            }
        }

        stage('Test') {
            steps {
                echo 'Testing the application...'
                sh 'docker image inspect my-docker-app'
            }
        }

        stage('Deploy') {
            steps {
                echo 'Deploying the application...'
                sh 'docker stop my-container || true'
                sh 'docker rm my-container || true'
                sh 'docker run -d -p 5000:5000 --name my-container my-docker-app'
            }
        }
    }
}
```



Jenkins

Docker-Pipeline

Configure



Configure



General



Source Code Management



Triggers



Environment



Build Steps



Post-build Actions

General

Enabled



Description

Plain text [Preview](#)



Discard old builds ?



GitHub project

Save

Apply



Jenkins

/ Docker-Pipeline / Configure



Advanced

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/ayushi2323/Devops-lab

Credentials ?

- none -



+ Add

Advanced

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

Save

Apply

```

ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git add Jenkinsfile
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Add Jenkins declarative pipeline"
[main 8e0a564] Add Jenkins declarative pipeline
1 file changed, 38 insertions(+)
create mode 100644 Jenkinsfile
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git push
Username for 'https://github.com': Docker-CICD
Password for 'https://Docker-CICD@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 791 bytes | 263.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
To https://github.com/ayushi2323/Devops-lab
5a81cb8..8e0a564 main -> main
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
14a5049451a2	jenkins/jenkins:lts	"/usr/bin/tini -- /u..."	About an hour ago	Up About an hour	0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp	jenkins
8f3b05f459fb	ae47b29b621b	"python main.py"	3 hours ago	Up 3 hours	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	my-container
7b82b6e69525	nginx	"/docker-entrypoint..."	3 hours ago	Up 3 hours	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	my-nginx